SkillClaw

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

Tagline: aggregate cross-user trajectories → evidence-driven skill updates
Source: arxiv.org/abs/2604.08377

Problem: static skills after deployment

Runtime failures & recoveries do not persist beyond individual sessions

Static skill hub

- • Users manually install skills
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- • Skills remain unchanged after deployment
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- • Solutions discovered in-session rarely persist
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- • Recurring tasks → repeated failure/recovery patterns across users

Evolving skill ecosystem

- • Collect interaction trajectories (action–feedback chains)
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- • Aggregate evidence across many users
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- • Diagnose failures & successes
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- • Propose skill mutations & verify
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- • Synchronize improved skills back to all

Existing approaches: memory-based methods are instance-tied and hard to generalize; skill-based libraries stay static after creation.

SkillClaw architecture overview

Multi-user interaction → session collection → agentic evolution → skill synchronization

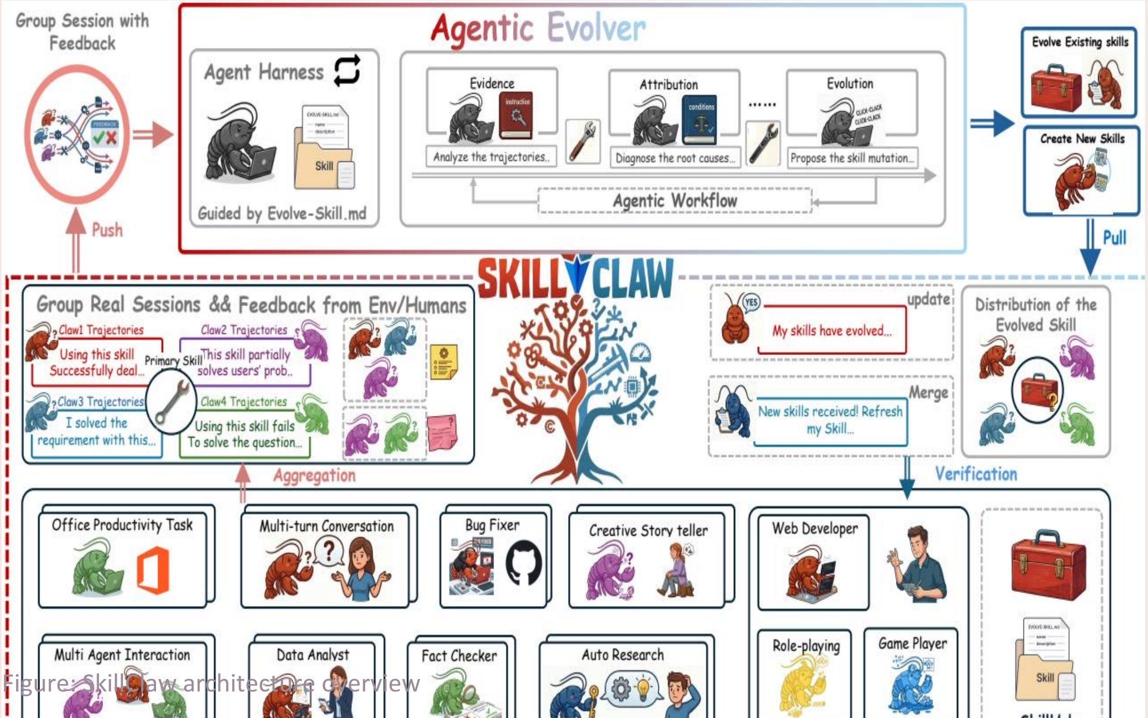


Figure: SkillClaw architecture overview

Four-stage closed loop

1) Evidence

2) Attribution

3) Evolution

Agentic evolver: open-ended reasoning over evidence

Three stages inside SkillClaw’s evolver (not predefined update rules)

1) Evidence

- Analyze aggregated trajectories
- Identify behavior patterns

2) Attribution

- Diagnose root causes
- Explain failures & successes
- Separate general signal vs

3) Evolution

- Propose skill mutations
- Refine existing skills or create new ones

After Evolution: verify → merge → synchronize

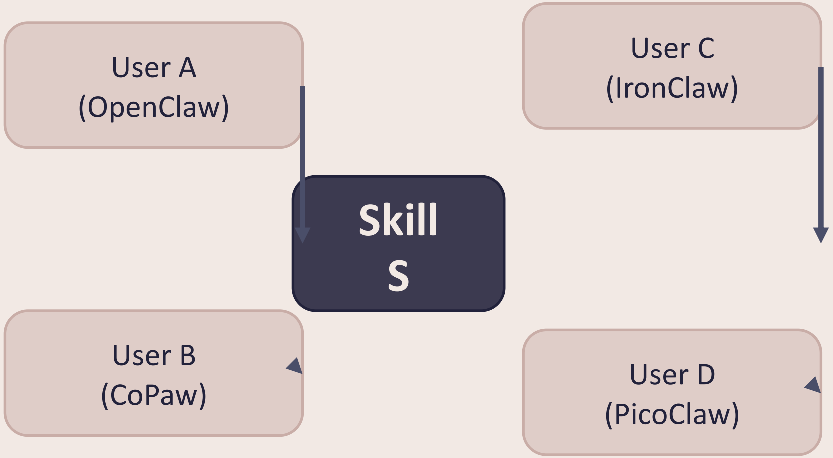
- Updated skills are verified before merging into the shared repository
- Synchronization pushes improved skills back to all agents
- Conservative verification prevents regressions

Agentic evolver stages from SkillClaw brief

Why multi-user aggregation matters

Diverse contexts reveal where a skill works vs. breaks

Aggregate trajectories by referenced skill



Cross-user aggregation rationale from SkillClaw brief

Key idea

- • Different users exercise the same skill under diverse contexts → complementary views
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- • Aggregation exposes a skill's behavioral boundary (where it works vs. breaks)
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- • A single user rarely provides enough signal to distinguish generalizable improvements

Evolution case study: multimodal creative task pipeline

Validator is conservative: 1/6 candidates accepted

Day	Date	Outcome	Rationale (brief)
1	Day 1	Accepted Accepted	validate-tmp-workspace-inputs: checks /tmp_workspace inputs & environment setup
2	Day 2	Rejected Rejected	multimodal input validation & output env init; kept current best
3	Day 3	Rejected Rejected	multimodal creative pipeline (extract from PDF/video/image); kept current best pool
4	Day 4	Rejected Rejected	added image classification + structured validation; kept current best pool
5	Day 5	Rejected Rejected	added per-file fail-fast validation; kept current best pool
6	Day 6	Rejected Rejected	extended PDF-to-poster/document-to-visual paths; not admitted

Case study table derived from SkillClaw brief

Evolution case study: git push auth-fallback

3/6 days yield accepted updates → refinement then plateau

Day	Date	Outcome	Rationale (brief)
1	Day 1	Accepted Accepted	safe fallback instead of blocking on push failure (added to Safety best pool)
2	Day 2	Accepted Accepted	unified patch/bundle filenames; reduced inconsistency
3	Day 3	Accepted Accepted	auth-alternative paths & secrets audit; fixed subshell pitfalls
4	Day 4	Accepted Accepted	same-pool retest; validator confirmed current pool as best
5	Day 5	Rejected Rejected	treated push hang as auth failure; no improvement
6	Day 6	Rejected Rejected	added identity config & filename consistency; no improvement; not

Case study table derived from SkillClaw brief

Takeaways & implications

What distinguishes SkillClaw for continuously improving agent systems

1) Collective evolution

Individual interactions contribute to a shared, continuously improving skill ecosystem.

2) Fully automatic

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention.

3) Agentic evolution paradigm

Updates are produced via open-ended reasoning over evidence, not predefined update rules.

Implications for agent systems

- Treat skills as living artifacts: updated via evidence, verified, and redistributed